

Instrumental Variables, 2SLS, and Instrument Design

Empirical Methods — Week 4

Teaching deck draft

Week 4 question

- When can an external source of variation identify a causal effect?
- What treatment margin does the instrument move?
- Why should the instrument affect outcomes only through treatment?
- For whom is the resulting estimate interpretable?

IV is a design argument before it is a two-stage regression.

Why IV is needed

$$Y_i = \beta D_i + X_i' \gamma + u_i$$

- OLS fails when D_i is correlated with unobserved determinants of Y_i .
- Endogeneity can come from ability, sorting, reverse causality, anticipation, or measurement error.
- IV asks for a source of variation Z_i that shifts treatment but is otherwise excluded from outcomes.
- The instrument's economic content defines the design.

First stage, reduced form, Wald

$$D_i = \pi Z_i + X_i' \delta + v_i \quad Y_i = \rho Z_i + X_i' \kappa + \varepsilon_i$$

$$\beta^{IV} = \frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]} = \frac{\text{Cov}(Y_i, Z_i)}{\text{Cov}(D_i, Z_i)}$$

- Numerator: instrument effect on outcome.
- Denominator: instrument effect on treatment.
- Ratio: effect of the treatment margin moved by the instrument.

$$W = [X, Z], \quad P_W = W(W'W)^{-1}W', \quad \hat{D} = P_W D$$

$$\hat{\theta}_{2SLS} = (\tilde{X}' P_W \tilde{X})^{-1} \tilde{X}' P_W Y, \quad \tilde{X} = [D, X]$$

- First stage keeps the treatment variation predicted by instruments and controls.
- Second stage relates outcomes to that instrument-predicted treatment.
- Projection notation does not create identification; the instrument design does.

Assumptions

Assumption	Meaning	Main design question
Relevance	Z shifts D	What treatment margin moves?
Independence	Z is as-good-as-random given X	What assignment or shock process creates variation?
Exclusion	Z affects Y only through D	What direct channels remain?
Monotonicity	no defiers	Could Z push some units away from treatment?
First-stage content	compliers are interpretable	Are they the margin the paper cares about?

$$\beta^{IV} = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0)]$$

- Compliers take treatment when exposed to the instrument and not otherwise.
- Always-takers and never-takers shape take-up levels, but not the Wald numerator-denominator margin.
- The estimate can be policy-relevant when compliers are the policy margin.
- It should not be reported as a universal ATE unless the design justifies that move.

Weak instruments and many instruments

$$F_{\text{excluded } Z} = \frac{(RSS_R - RSS_U)/q}{RSS_U/(n - k_U)}$$

- Weak first stages make 2SLS noisy, biased toward OLS, and specification-sensitive.
- Report first-stage coefficients, economic magnitude, and weak-IV diagnostics.
- Use weak-IV robust inference when the first stage is fragile.
- Many instruments can overfit the first stage and make interpretation opaque.
- Multiple instruments may combine different complier populations.

Instrument design gallery

Family	First-stage logic	Core threat
Policy rules	eligibility or legal exposure shifts treatment	cohort, seasonality, or rule channels
Distance/access	lower cost raises take-up	geography directly affects outcomes
Leniency leave-out	assigned decision-makers differ in propensities	decision-makers affect outcomes through other channels
Shift-share	shocks hit exposed units differently	endogenous shares or non-random shocks
Historical legacy	past conditions predict current treatment	many persistent pathways to current outcomes

Every family needs its own first-stage story, exclusion argument, and inference level.

Shift-share instruments

$$Z_\ell = \sum_s w_{\ell s} g_s$$

- Shares $w_{\ell s}$ measure exposure; shocks g_s create differential predicted treatment.
- Modern practice asks whether identification comes from quasi-random shocks or as-good-as-random shares.
- Endogenous shares can proxy local trends, institutions, or industry composition.
- Inference should reflect the shock structure, not only geographic clustering.
- Report dominant shocks, leave-out choices, exposure balance, and shock-level sensitivity.

Leniency leave-out instruments

$$Z_{ij} = \frac{1}{N_j - 1} \sum_{k \neq i: J_k = j} D_k$$

- The leave-out measure avoids using case i to predict its own treatment.
- Identification requires quasi-random assignment to decision-makers within cells.
- Balance checks should be done inside the assignment cells that define randomization.
- Exclusion can fail if lenient decision-makers change information, processing time, stigma, or other treatments.
- The effect is local to marginal cases moved by decision-maker style.

Historical, distance, and policy instruments

- Policy rules: compelling when the rule is sharp, binding, and isolated from other channels.
- Distance/access: useful when access cost is the treatment margin; fragile when geography carries opportunity or family background.
- Historical instruments: powerful first stages, but persistent direct channels are the central threat.
- The implementation task is to name the mechanism, show first-stage content, and stress-test exclusion.

Reproduce

- Compulsory-schooling synthetic IV
- First stage
- Reduced form
- Wald and 2SLS

Diagnose

- Partial first-stage F
- Balance by instrument
- Complier profile
- Exclusion memo

Transfer

- Shift-share construction
- Shock-level diagnostics
- Endogenous shares
- Leniency comparison

- IV estimates the treatment margin induced by the instrument.
- The first stage defines the compliers and the economic interpretation.
- Exclusion is the central substantive assumption and is not directly testable.
- Weak and many instruments can make formal 2SLS output misleading.
- Instrument families are not interchangeable; each has distinct implementation caveats.
- A credible IV paper is a theory of variation before it is a regression table.